

Quantifying leaf vein traits

叶脉性状量化协议

Bilingual English-Chinese translation with barley-specific practical notes. Source: PrometheusWiki protocol uploaded as “ 2013 Sack & Scoffoni, Vein Structure(1).pdf ”.

说明：本文为中英文对照翻译和校对版。原文中的叶脉等级示意图已保留，并在图旁补充了中文对照标签，尽量不遮挡英文原标注。实际实验涉及甲醛、冰醋酸、NaOH、漂白剂、乙醇和染色剂，应遵守所在实验室的SDS、风险评估、PPE、废液处理和机构审批要求。

Overview 概述

EN: This protocol explains how to clear and image leaves and to make basic measurement of venation traits. Important vein traits include vein densities (length/area) and diameters for all vein orders, and the number of free-ending veins per area.	中文：本协议说明如何对叶片进行透明化处理和成像，并对叶脉性状进行基本测量。重要的叶脉性状包括各级叶脉的叶脉密度（长度/面积）、各级叶脉直径，以及单位面积内自由末端叶脉的数量。
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Background 背景

EN: This protocol outlines how to obtain the traits used to quantify structure and to relate to the functions of leaf venation architecture. Leaf venation architecture has common functions across plant species, serving for mechanical support (Niklas, 1999), sugar and hormone transport in the phloem (Kehr and Buhtz, 2008), and, via the xylem, the replacement of water lost to transpiration during photosynthesis (Sack and Holbrook, 2006). However, venation architecture is highly diverse across species (Uhl and Mosbrugger, 1999; Roth-Nebelsick et al., 2001; Sack and Frole, 2006; Ellis et al., 2009; Brodribb et al., 2010).	中文：本协议概述了如何获得用于量化叶片结构的性状，并将这些性状与叶脉结构功能联系起来。叶脉结构在不同植物物种中具有共同功能，包括提供机械支撑（Niklas, 1999）、通过韧皮部运输糖和激素（Kehr and Buhtz, 2008），以及通过木质部补充光合作用蒸腾过程中损失的水分（Sack and Holbrook, 2006）。然而，不同物种之间的叶脉结构差异很大（Uhl and Mosbrugger, 1999; Roth-Nebelsick et al., 2001; Sack and Frole, 2006; Ellis et al., 2009; Brodribb et al., 2010）。
EN: In dicotyledons, the leaf venation system typically consists of three orders of major veins and up to five higher orders of minor veins embedded in the mesophyll, with the vein orders arranged in a hierarchy; lower order veins are larger in diameter, with greater xylem conduit numbers and sizes, whereas higher order veins have greater length per area (vein density; Sack and Holbrook, 2006; McKown et al., 2010).	中文：在双子叶植物中，叶脉系统通常由三级主脉以及最多五级嵌入叶肉中的更高级细脉组成，各级叶脉按层级结构排列。较低级别的叶脉直径更大，木质部导管数量更多且尺寸更大；而较高级别的叶脉单位面积长度更高，即叶脉密度更高（Sack and Holbrook, 2006; McKown et al., 2010）。
EN: Total leaf vein density has been shown to correlate with maximum hydraulic conductance and photosynthetic rate per area across species (Sack and Frole, 2006; Brodribb et al., 2007) and tends to be higher for species growing in high light. Major vein density has been found to play a role in determining the damage tolerance of the vein system, and in leaf drought tolerance (Sack et al., 2008; Scoffoni et al., 2011).	中文：研究表明，跨物种比较时，总叶脉密度与最大水力导度以及单位面积光合速率相关（Sack and Frole, 2006; Brodribb et al., 2007），并且在高光环境中生长的物种通常具有更高的总叶脉密度。主脉密度也被发现会影响叶脉系统对损伤的耐受性，以及叶片的抗旱能力（Sack et al., 2008; Scoffoni et al., 2011）。

Materials/Equipment 材料与设备

EN: Chemicals: Aqueous formaldehyde, glacial acetic acid solution, EtOH, NaOH, Bleach.	中文：化学品：甲醛水溶液、冰醋酸溶液、乙醇（EtOH）、氢氧化钠（NaOH）、漂白剂。
EN: Stains: Safranin, Fast green.	中文：染色剂：番红（Safranin）、固绿（Fast green）。

EN: Equipment: Flatbed scanner, light microscope, ImageJ software (National Institutes of Health; free online, http://rsbweb.nih.gov/ij/).	中文：设备：平板扫描仪、光学显微镜、ImageJ 软件（美国国立卫生研究院，National Institutes of Health；可在线免费获取： http://rsbweb.nih.gov/ij/ ）。
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Units, terms, definitions 单位、术语与定义

EN: Vein density = Vein length per unit area (mm mm^{-2}). Can be measured for veins of each order, or summed for major or minor veins or the whole system (see below). Total vein density correlates well with the average distance between veins (a.k.a. "interveinal distance"; see Uhl and Mosbrugger, 1999), and with areoles per area.	中文：叶脉密度 = 单位面积内的叶脉长度，单位为 mm m m^{-2} 。可以分别测量各级叶脉，也可以汇总为主脉、细脉或整个叶脉系统的密度（见下文）。总叶脉密度与叶脉之间的平均距离，即“脉间距”（interveinal distance；见 Uhl and Mosbrugger, 1999）关系密切，也与单位面积内的网眼（areoles）数量相关。
EN: Free vein endings per area = Number of free vein endings per unit area (mm^{-2}).	中文：单位面积自由末端叶脉数 = 单位面积内自由末端叶脉的数量，单位为 mm^{-2} 。
EN: Major vein density = Sum of vein densities for 1°, 2° and 3° veins (mm mm^{-2}).	中文：主脉密度 = 一级、二级和三级叶脉密度之和，单位为 mm mm^{-2} 。
EN: Minor vein densities = Sum of vein densities for 4° veins and higher (mm mm^{-2}).	中文：细脉密度 = 四级及以上叶脉密度之和，单位为 mm mm^{-2} 。
EN: Vein diameter = typical cross-sectional diameter of a central vein segment for a vein of a given order (mm).	中文：叶脉直径 = 某一叶脉等级中典型中央叶脉段的横截面直径，单位为 mm。

Procedure 实验步骤

EN: Leaves can be preserved in formalin-acetic acid solution (37% aqueous formaldehyde solution, 50% ethanol, and 13% glacial acetic acid solution) until they are ready for clearing.	中文：叶片可保存于福尔马林-醋酸固定液中，直到准备进行透明化处理。该固定液由 37% 甲醛水溶液、50% 乙醇和 13% 冰醋酸溶液组成。
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Leaf clearing, simplified from Berlyn and Miksche, 1976 叶片透明化处理，简化自 Berlyn and Miksche, 1976

EN: It may take several attempts for good results for given species. The method given below is flexible (variable times for given steps are presented) and so if an attempt fails, modify given steps as necessary for improved results.	中文：对于特定物种，可能需要多次尝试才能获得较好结果。下面的方法具有一定灵活性，各步骤给出了可变处理时间。因此，如果某次尝试失败，可根据需要调整相关步骤，以获得更好的结果。
EN: Keep notes on your attempts and you will eventually arrive at a functional process for leaves of given species. Enjoy!	中文：请记录每次尝试的条件和结果，最终你会为特定物种的叶片摸索出可行的处理流程。祝实验顺利！
EN: Chemically clear leaves using 5% NaOH in aqueous solution (let the leaf sit in it for a few hours to a few days depending on the species), or use a 2.5% solution for delicate tissues.	中文：使用 5% NaOH 水溶液对叶片进行化学透明化处理。根据物种不同，叶片可在溶液中放置数小时至数天。对于较脆弱的组织，可使用 2.5% NaOH 溶液。
EN: When leaves appear transparent and softened, vacuum off the NaOH solution and rinse several times in H ₂ O.	中文：当叶片变得透明并软化时，抽去 NaOH 溶液，并用 H ₂ O 冲洗数次。

EN: If a little color remains in the leaf, rinse out the H ₂ O and pour a 50% bleach solution on the leaf for 10-20 s. Rinse twice with H ₂ O and let sit for at least 1 hour.	中文：如果叶片中仍残留少量颜色，先去除 H ₂ O，然后向叶片上加入 50% 漂白剂溶液处理 10 到 20 秒。随后用 H ₂ O 冲洗两次，并静置至少 1 小时。
EN: Bring leaves into alcohol again using EtOH dilution series (30%, 50%, 70%, 100%). Each change needs no more than 5 mins. Vacuum off liquid.	中文：通过 EtOH 梯度系列将叶片重新转入乙醇中，浓度依次为 30%、50%、70%、100%。每次换液不需要超过 5 分钟。随后抽去液体。
EN: After the 100% EtOH stage, cover the leaf in 1% safranin (1 g safranin/100 ml of 100% EtOH). The leaf should sit 2-30 minutes depending on the species. Gently rinse with 100% EtOH and vacuum off liquid.	中文：完成 100% EtOH 步骤后，用 1% 番红溶液覆盖叶片，配方为 1 g 番红加入 100 ml 100% EtOH。根据物种不同，叶片应浸泡 2 到 30 分钟。随后用 100% EtOH 轻轻冲洗，并抽去液体。
EN: Cover leaf in 1% fast green (1 g fast green/100 ml of 100% EtOH) and leave for a few seconds. Gently rinse with 100% EtOH and vacuum off liquid.	中文：用 1% 固绿溶液覆盖叶片，配方为 1 g 固绿加入 100 ml 100% EtOH，并放置数秒。随后用 100% EtOH 轻轻冲洗，并抽去液体。
EN: Bring leaf back to water in the reverse dilution series. The leaf may look overstained, but once it is in water stage, the excess dye will come out and should leave dark-stained veins (see attached image). Skipping these steps and going straight to water can mean getting an overstained leaf or a damaged leaf - take care as your tissues are soft. Once in the H ₂ O stage, if you find your leaf is understained, bring back to 100% EtOH and repeat the stain procedure.	中文：按照反向稀释梯度将叶片带回水相。叶片一开始可能看起来染色过深，但进入水相后，多余染料会逐渐脱出，理论上会留下深染的叶脉（见原文附图）。如果跳过这些步骤直接转入水中，可能会导致叶片过度染色或损伤。此时组织较软，需要小心操作。进入 H ₂ O 阶段后，如果发现叶片染色不足，可将其重新转回 100% EtOH，并重复染色步骤。
EN: Once in the H ₂ O stage, leaves can be mounted in H ₂ O for vein imaging and analysis.	中文：叶片进入 H ₂ O 阶段后，可用 H ₂ O 封片，用于叶脉成像和分析。

Vein imaging 叶脉成像

EN: The leaf is mounted with water in transparency film (CG5000; 3M Visual Systems Division).	中文：将叶片用水装片于透明胶片中。原文示例为 CG5000, 3M Visual Systems Division。
EN: Rules should be made for distinguishing vein orders in your leaves (for excellent advice, see Ellis et al., 2009). There is a level of subjectivity and uncertainty in making these distinctions, so attempt to be rigorous and consistent, and keep notes on your rules for distinguishing vein orders (see image below for an example of vein order determination on <i>Heteromeles arbutifolia</i>).	中文：应为待测叶片建立区分不同叶脉等级的规则，相关建议可参考 Ellis et al., 2009。区分叶脉等级时存在一定主观性和不确定性，因此应尽量严格且保持一致，并记录自己用于区分叶脉等级的规则。原文下方图像提供了 <i>Heteromeles arbutifolia</i> 叶脉等级判定的示例。
EN: First, the leaf is scanned to allow measurement of the density of 1°, 2° and 3° veins, and the diameter of 1° vein(s). Next, the leaf is imaged under the light microscope to allow measurement of higher order veins.	中文：首先扫描整个叶片，用于测量一级、二级和三级叶脉的密度，以及一级叶脉的直径。随后在光学显微镜下对叶片成像，用于测量更高级别的叶脉。
EN: The whole leaf is scanned (flatbed scanner; Canon Scan Lide 90; 1,200 pixels per inch).	中文：扫描整片叶片。原文使用平板扫描仪 Canon Scan Lide 90，分辨率为每英寸 1,200 像素。
EN: Using a light microscope (DMRB; Leica Microsystems) with a 5x or 10x objective and digital camera (14.2 Color Mosaic; Diagnostic Instruments), image three rectangles (at least 1.5 mm ²) centrally in the top, middle and bottom thirds of the leaf.	中文：使用光学显微镜（原文为 Leica Microsystems 的 DMRB），配合 5 倍或 10 倍物镜以及数字相机（原文为 Diagnostic Instruments 的 14.2 Color Mosaic），在叶片上部、中部和下部三等分区域的中央位置分别拍摄三个矩形区域，每个区域至少 1.5 mm ² 。

EN: A good practice is to make sure that the vein at the top of the image is part of a 2° or 3° vein depending on the species (see below), so you can orient yourself to the vein orders in the image later.	中文：一个较好的做法是确保图像顶部的叶脉属于二级或三级叶脉，具体取决于物种。这样后续分析图像时，可以更容易判断图中叶脉等级的位置关系。
EN: Image two areas focusing on a typical central 2° vein.	中文：另外拍摄两个区域，重点聚焦于典型的中央二级叶脉。

Vein quantification 叶脉量化

Quantification of whole leaf features 整片叶片特征量化

EN: Using ImageJ software (National Institutes of Health) measure leaf area, perimeter, length and width.	中文：使用 ImageJ 软件（National Institutes of Health）测量叶面积、周长、长度和宽度。
EN: Two indices of leaf shape can be calculated. The length:width ratio and the perimeter ² :area ratio which is a size-independent index of edge relative to size (Sack et al. 2003).	中文：可以计算两个叶形指数：长宽比，以及周长平方/面积比（perimeter ² :area ratio）。后者是一个与大小无关的指数，用于表示叶片边缘相对于叶片大小的程度（Sack et al. 2003）。

Quantification of major veins 主脉量化

EN: Using ImageJ software, measure the total lengths of 1° and 2° veins.	中文：使用 ImageJ 软件测量一级和二级叶脉的总长度。
EN: 3° vein lengths are measured in three rectangles per leaf size (area of rectangle would depend on leaf size) located centrally in the top, middle and bottom thirds of half the leaf.	中文：三级叶脉长度在每片叶的三个矩形区域中测量，矩形面积取决于叶片大小。这三个矩形位于半片叶的上部、中部和下部三等分区域的中央位置。
EN: Measure the 1° vein diameter, excluding the bundle sheath, by averaging two measurements done centrally in the 1° vein.	中文：测量一级叶脉直径时，应排除维管束鞘。在一级叶脉中央位置进行两次测量，并取平均值。
EN: Vein densities for each order are calculated as the ratio of vein length/leaf area for 1° and 2° veins, and as 3° length / rectangle area for 3° veins.	中文：各级叶脉密度的计算方法为：一级和二级叶脉按叶脉长度/叶面积计算；三级叶脉按三级叶脉长度/矩形区域面积计算。
EN: Major vein density is calculated as the sum of the 1°, 2° and 3° vein densities.	中文：主脉密度计算为一级、二级和三级叶脉密度之和。
EN: From the light microscope images of 2° veins, measure the vein diameter centrally twice.	中文：在二级叶脉的光学显微镜图像中，在中央位置测量叶脉直径两次。

Quantification of minor veins 细脉量化

EN: From each image at the top, middle, bottom part of the leaf, measure:	中文：对叶片上部、中部和下部各图像分别测量以下指标：
EN: Total image area	中文：总图像面积
EN: Area occupied by 2° veins	中文：二级叶脉所占面积
EN: Area occupied by 3° veins	中文：三级叶脉所占面积

EN: 2° vein length	中文：二级叶脉长度
EN: 3° vein length	中文：三级叶脉长度
EN: Total length of 4° veins and higher	中文：四级及以上叶脉的总长度
EN: Number of free vein endings	中文：自由末端叶脉数量
EN: Vein diameters centrally for two segments of each vein order (3°, 4°, 5°, 6° and 7° if existent)	中文：各叶脉等级两个叶脉段中央位置的叶脉直径，包括三级、四级、五级、六级和存在时的七级叶脉
EN: Minor vein density is calculated as:	中文：细脉密度计算为：

(sum of the vein lengths of 4° veins and higher) / (total image area - area occupied by 2° veins)

(四级及以上叶脉长度之和) / (总图像面积 - 二级叶脉所占面积)

EN: If necessary, you can also remove the area occupied by 3° veins from the denominator, but this might not be practical if there are many such veins in the image, and the area occupied in the image relatively very small.	中文：如有必要，也可以在分母中去除三级叶脉所占面积。但如果图像中三级叶脉很多，且其所占面积相对很小，这样做可能并不实际。
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EN: Total vein density can be calculated as:	中文：总叶脉密度可计算为：
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(sum of the vein lengths of all orders of veins) / (total image area)

(所有等级叶脉长度之和) / (总图像面积)

EN: It would be best to calculate this for the image area without 1° or 2° veins. Alternatively, total vein density can be calculated as major vein density plus minor vein density, or ideally as major vein density plus minor vein density x (1 - fraction of leaf area occupied by major veins), where the fraction can be determined as the sum of diameter x vein density for 1°, 2° and 3° veins. Values determined these ways tend to be extremely highly correlated.	中文：最好在不包含一级或二级叶脉的图像区域中计算总叶脉密度。另一种方法是将总叶脉密度计算为主脉密度加细脉密度。更理想的计算方法是：主脉密度 + 细脉密度 x (1 - 主脉占叶面积的比例)。其中，主脉占叶面积的比例可通过一级、二级和三级叶脉的“直径 x 叶脉密度”之和来确定。用这些方法得到的数值通常高度相关。
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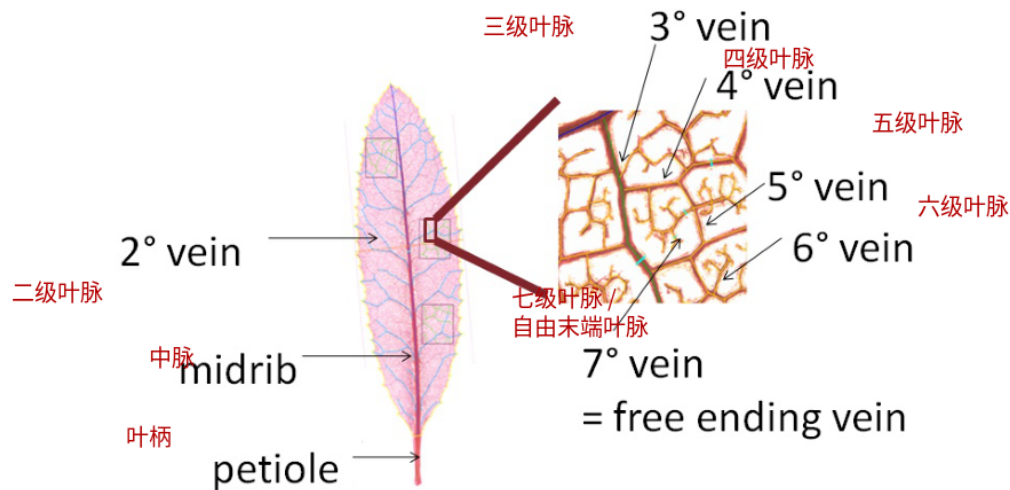
EN: Free vein endings per area is calculated as:	中文：单位面积自由末端叶脉数计算为：
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(number of free vein endings) / (total image area - 2° vein area)

(自由末端叶脉数量) / (总图像面积 - 二级叶脉面积)

Original figure note 原文图示说明

EN: The original page 3 diagram labels the midrib, petiole, 2° vein, 3° vein, 4° vein, 5° vein, 6° vein, 7° vein, and free-ending vein.	中文：原文第3页图中标出了中脉 (midrib)、叶柄 (petiole)、二级叶脉、三级叶脉、四级叶脉、五级叶脉、六级叶脉、七级叶脉，以及自由末端叶脉。
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EN: Scoffoni C, Rawls M, McKown AD, Cochard H, Sack L (2011) Decline of leaf hydraulic conductance with dehydration: relationship to leaf size and venation architecture. <i>Plant Physiology</i> : in press.	中文：Scoffoni C, Rawls M, McKown AD, Cochard H, Sack L (2011) 脱水过程中叶片水力导度下降：与叶片大小和叶脉结构的关系。 <i>Plant Physiology</i> ：印刷中。
EN: Uhl D, Mosbrugger V (1999) Leaf venation density as a climate and environmental proxy: a critical review and new data. <i>Palaeogeography Palaeoclimatology Palaeoecology</i> 149, 15-26.	中文：Uhl D, Mosbrugger V (1999) 叶脉密度作为气候和环境代用指标：批判性综述与新数据。 <i>Palaeogeography Palaeoclimatology Palaeoecology</i> 149, 15-26。
EN: The original document is available at http://prometheus.wiki.org/tiki-index.php?page=Quantifying+leaf+vein+traits	中文：原始文件可通过以下地址获取： http://prometheus.wiki.org/tiki-index.php?page=Quantifying+leaf+vein+traits

Appendix 1. Preparation checklist 实验前准备清单

EN: Before starting the protocol, confirm your leaf species, sample number, imaging plan, and whether you need preservation before clearing.	中文：开始实验前，请先确认叶片物种、样本数量、成像计划，以及是否需要在透明化前先进行固定保存。
EN: Recommended planning items:	中文：建议提前准备以下事项：
EN: Leaf samples: fresh leaves or preserved leaves, with clear sample labels and replicate plan.	中文：叶片样品：新鲜叶片或已固定保存的叶片，并做好清晰编号与重复设计。
EN: Containers: small glass vials, staining dishes, Petri dishes, or chemical resistant tubes for each step.	中文：容器：用于各步骤的小玻璃瓶、染色皿、培养皿或耐化学品离心管。
EN: Handling tools: forceps, transfer pipettes or Pasteur pipettes, wash bottle, timer, and absorbent tissue.	中文：操作工具：镊子、转移滴管或巴斯德吸管、洗瓶、计时器和吸水纸。
EN: Water source: distilled or deionized water for rinsing and mounting.	中文：水源：用于冲洗和制片的蒸馏水或去离子水。
EN: Mounting material: transparency film such as CG5000, microscope slides if needed, and covers if your local method requires them.	中文：制片材料：透明胶片，如 CG5000；如本地方法需要，也可准备载玻片和盖玻片。
EN: Imaging equipment: flatbed scanner, light microscope, digital camera, and a computer with ImageJ installed.	中文：成像设备：平板扫描仪、光学显微镜、数码相机，以及已安装 ImageJ 的电脑。
EN: Vacuum removal or aspiration setup if your lab uses vacuum to remove liquid between steps.	中文：抽液或真空装置：如果实验室采用抽真空或吸液方式换液，需要提前准备。
EN: PPE and approvals: lab coat, gloves, eye protection, and any required risk assessment or supervisor approval.	中文：个人防护与审批：实验服、手套、护目镜，以及所需的风险评估或导师审批。

Appendix 2. Chemical checklist and suggested solutions 化学品清单与建议配制

EN: Core chemicals named in the protocol:	中文：该 protocol 明确提到的核心化学品包括：
EN: Aqueous formaldehyde	中文：甲醛水溶液
EN: Glacial acetic acid solution	中文：冰醋酸溶液
EN: Ethanol, including 30%, 50%, 70%, and 100% EtOH series	中文：乙醇，包括 30%、50%、70% 和 100% EtOH 梯度
EN: Sodium hydroxide, typically 5% NaOH, or 2.5% for delicate tissues	中文：氢氧化钠，通常为 5% NaOH，脆弱组织可用 2.5%
EN: Bleach, used here as a 50% bleach solution	中文：漂白剂，此 protocol 中使用 50% 漂白剂溶液
EN: Safranin stain, prepared here as 1% in 100% EtOH, 1 g per 100 ml	中文：番红染色液，此处为 1% 番红的 100% EtOH 溶液，即 1 g/100 ml
EN: Fast green stain, prepared here as 1% in 100% EtOH, 1 g per 100 ml	中文：固绿染色液，此处为 1% 固绿的 100% EtOH 溶液，即 1 g/100 ml
EN: Distilled or deionized water	中文：蒸馏水或去离子水
EN: Solutions explicitly described in the protocol:	中文：protocol 中明确写出的溶液包括：
EN: Formalin acetic acid storage solution: 37% aqueous formaldehyde solution, 50% ethanol, and 13% glacial acetic acid solution.	中文：福尔马林-醋酸保存液：37% 甲醛水溶液、50% 乙醇和 13% 冰醋酸溶液。
EN: Clearing solution: 5% NaOH in water, or 2.5% NaOH for delicate tissues.	中文：透明化溶液：5% NaOH 水溶液，脆弱组织可用 2.5% NaOH。
EN: Decolourising step if needed: 50% bleach solution for 10 to 20 seconds.	中文：必要时的脱色步骤：50% 漂白剂溶液，处理 10 到 20 秒。
EN: Alcohol series: 30%, 50%, 70%, 100% EtOH, each change up to 5 minutes.	中文：乙醇梯度：30%、50%、70%、100% EtOH，每次换液最多约 5 分钟。
EN: Safranin staining: 1% safranin in 100% EtOH, 2 to 30 minutes depending on species.	中文：番红染色：1% 番红的 100% EtOH 溶液，根据物种染色 2 到 30 分钟。
EN: Fast green staining: 1% fast green in 100% EtOH, a few seconds only.	中文：固绿染色：1% 固绿的 100% EtOH 溶液，仅染数秒。
EN: Reverse alcohol series back to water before mounting.	中文：染色后通过反向乙醇梯度回到水相，再进行制片。
EN: Practical note	中文：实际准备提示
EN: It is wise to prepare small working volumes first, because different species may require optimisation and repeated trials. Also check your laboratory SDS, local waste disposal rules, and whether formaldehyde and bleach must be handled in a fume hood.	中文：建议先配制小体积工作液，因为不同物种可能需要优化并重复试验。另外，请确认实验室 SDS、本地废液处理规则，以及甲醛和漂白剂是否必须在通风橱中操作。

Appendix 3. Quick shopping checklist 超简版购物清单

EN: This page is designed as a simple buy and prepare list before you start.	中文：这一页是给你真正开做前用的简化清单，可以直接照着买、照着准备。
EN: Must have, chemicals	中文：必备化学品
EN: 37% aqueous formaldehyde solution	中文：37% 甲醛水溶液
EN: Glacial acetic acid	中文：冰醋酸
EN: Ethanol, enough to prepare 30%, 50%, 70%, and 100% series	中文：乙醇，足够配 30%、50%、70% 和 100% 梯度
EN: NaOH pellets or ready made NaOH solution	中文：NaOH 固体或现成 NaOH 溶液
EN: Household bleach or laboratory bleach source	中文：漂白剂或实验室可用漂白液来源
EN: Safranin	中文：番红
EN: Fast green	中文：固绿
EN: Distilled or deionized water	中文：蒸馏水或去离子水
EN: Must have, consumables and tools	中文：必备耗材与工具
EN: Sample tubes, vials, or small glass containers	中文：样品管、小玻璃瓶或小容器
EN: Petri dishes or staining dishes	中文：培养皿或染色皿
EN: Forceps and transfer pipettes	中文：镊子和转移滴管
EN: Wash bottle	中文：洗瓶
EN: Labels and waterproof marker	中文：标签和防水记号笔
EN: Timer	中文：计时器
EN: Absorbent tissue or paper towel	中文：吸水纸或纸巾
EN: Transparency film such as CG5000	中文：透明胶片，例如 CG5000
EN: PPE: gloves, lab coat, eye protection	中文：PPE：手套、实验服、护目镜
EN: Must have, imaging and analysis	中文：必备成像与分析工具
EN: Flatbed scanner	中文：平板扫描仪
EN: Light microscope with 5x or 10x objective	中文：配有 5 倍或 10 倍物镜的光学显微镜
EN: Digital camera connected to microscope if available	中文：如有条件，显微镜配套数码相机
EN: Computer with ImageJ installed	中文：安装好 ImageJ 的电脑

EN: Nice to have	中文：可选但很有帮助
EN: Fume hood access	中文：通风橱使用条件
EN: Vacuum aspiration setup	中文：抽液或真空吸液装置
EN: Extra small volume bottles for trial runs	中文：用于预实验的小体积试剂瓶
EN: Notebook or spreadsheet for recording staining times by species	中文：用于记录不同物种处理时间的实验记录本或表格
EN: Small batch suggestion	中文：小批量准备建议
EN: For your first trial, prepare only small working volumes of NaOH, bleach, safranin, and fast green, because staining and clearing times may need optimisation.	中文：第一次预实验时，建议只少量配制 NaOH、漂白剂、番红和固绿工作液，因为透明化和染色时间往往需要优化。

Appendix 4. Practical notes for barley leaf structure 针对大麦叶片结构的实操建议

EN: The original protocol is general. The following notes are practical suggestions for barley leaves and are added for convenience.	中文：原始 protocol 是通用型的。下面内容是针对大麦叶片的实操建议，属于补充说明，方便你开始预实验。
EN: Important barley-specific note: the original protocol and figure are mainly framed around dicot reticulate venation. Barley is a monocot grass, so its leaf blade mainly shows parallel venation. For barley, it is usually more appropriate to focus on longitudinal vein density, interveinal distance, vascular bundle or vein diameter, and consistent imaging positions, rather than forcing all dicot-style vein orders and free-ending vein metrics.	中文：大麦专门提示：原 protocol 和图示主要以双子叶植物的网状叶脉为背景。大麦是单子叶禾本科植物，叶片主要表现为平行脉。因此，大麦实验中通常更适合重点关注纵向叶脉密度、脉间距、维管束或叶脉直径，以及固定一致的成像位置，而不是强行套用所有双子叶式叶脉等级和自由末端叶脉指标。
EN: 1. Leaf choice	中文：1. 叶片选择
EN: Choose healthy, fully expanded leaves with clear labels for genotype, treatment, and replicate. Avoid severely senescent, heavily curled, or mechanically damaged leaves for your first optimisation round.	中文：建议选择健康、充分展开的叶片，并清楚标注基因型、处理和重复。第一次优化时，尽量不要用严重衰老、强烈卷曲或机械损伤明显的叶片。
EN: 2. Why barley may need extra care	中文：2. 为什么大麦需要更小心
EN: Barley leaves are narrow monocot grass leaves with mainly parallel venation. After chemical clearing, they may become too soft, tear during transfers, or become overstained if steps are too long.	中文：大麦叶片属于狭长的单子叶禾本科叶片，叶脉以平行脉为主。经过化学透明化后，叶片可能变得较软，转移时容易破损，处理时间过长也可能染色过深。
EN: 3. Steps most likely to cause trouble	中文：3. 最容易出问题的步骤
EN: NaOH clearing: too long can make the leaf overly soft or damaged.	中文：NaOH 透明化：时间过长会让叶片过软，甚至受损。
EN: Bleach step: too strong or too long can overdecolourise or damage tissue.	中文：漂白步骤：浓度过高或时间过长，容易过度脱色或损伤组织。
EN: Safranin staining: too long can make the whole leaf look too dark.	中文：番红染色：时间过长会导致整片叶子颜色过深。
EN: Fast green staining: even a few extra seconds may strongly affect colour balance.	中文：固绿染色：哪怕多几秒，也可能明显改变染色平衡。
EN: Transfers between solutions: soft leaves may fold, tear, or stick to tools.	中文：换液转移：变软后的叶片容易折叠、破损，或粘在工具上。
EN: 4. Sensible first trial strategy	中文：4. 合理的第一次预实验策略
EN: Start conservatively. For barley, it is sensible to trial a small number of leaves first and to test shorter treatment windows before scaling up.	中文：建议先保守一点。对于大麦，最好先用少量叶片做预实验，并优先测试较短的处理时间，再决定是否放大实验。
EN: You may begin with the lower strength option for delicate tissue, such as 2.5% NaOH, and only move to 5% NaOH if clearing is insufficient.	中文：可以先从较温和的条件开始，例如对脆弱组织先尝试 2.5% NaOH。如果透明化不够，再提高到 5% NaOH。

EN: 5. Chemicals worth preparing in small amounts first	中文：5. 建议先少量准备的化学品
EN: 2.5% and 5% NaOH working solutions	中文：2.5% 和 5% NaOH 工作液
EN: 50% bleach working solution	中文：50% 漂白剂工作液
EN: 1% safranin in 100% EtOH	中文：1% 番红的 100% EtOH 溶液
EN: 1% fast green in 100% EtOH	中文：1% 固绿的 100% EtOH 溶液
EN: Small volumes of each EtOH concentration in the dilution series	中文：乙醇梯度各浓度的小体积工作液
EN: 6. Imaging suggestions	中文：6. 成像建议
EN: Because barley venation is mainly parallel and can be visually repetitive, keep the imaging locations consistent across leaves. It helps to define in advance which part of the blade you will image, for example upper, middle, and lower sections, while avoiding damaged edges.	中文：由于大麦叶脉以平行脉为主，在视觉上可能较为重复，建议在不同叶片之间保持一致的成像位置。最好提前规定拍摄叶片的哪个部位，例如上部、中部和下部，同时避免边缘破损区域。
EN: 7. Record keeping	中文：7. 记录建议
EN: For each trial, record leaf age or position, solution strengths, actual exposure times, whether the leaf became too soft, and whether the final image quality was acceptable. These notes will save a lot of time during optimisation.	中文：每次预实验都建议记录叶龄或叶位、溶液浓度、实际处理时间、叶片是否过软，以及最终图像质量是否可接受。这些记录会在后续优化中非常省时间。
EN: 8. Final reminder	中文：8. 最后提醒
EN: This protocol involves hazardous chemicals. Before starting, make sure your supervisor or laboratory has approved the work, and always follow local SDS, PPE, and waste disposal requirements.	中文：该 protocol 涉及危险化学品。开始前请确保导师或实验室已批准该实验，并始终遵守本地 SDS、PPE 和废液处理要求。

Appendix 5. Workflow ordered solution prep list 按实验流程排序的配液清单

EN: This page is a practical working checklist. It is not part of the original protocol, but it can help you prepare in a logical order before starting the experiment.	中文：这一页是实操型工作清单，不属于原始 protocol 正文，但可以帮助你正式开始前按更合理的顺序准备。
EN: 1. Before making solutions ■■■■■■	中文：1. Before making solutions, 开始配液前
EN: Label all tubes, bottles, dishes, and waste containers first. Also decide how many leaves you will process in your first trial.	中文：先给所有离心管、试剂瓶、培养皿和废液容器贴好标签。同时先决定第一次预实验要处理多少片叶子。
EN: 2. Prepare clean water ■■■■■■	中文：2. Prepare clean water, 先准备纯水
EN: Prepare enough distilled or deionized water for rinsing, dilution, and mounting. This is the base solution you will use throughout the workflow.	中文：先准备足够的蒸馏水或去离子水，用于冲洗、稀释和制片。这是整个流程都会用到的基础液体。
EN: 3. Prepare the ethanol series ■■■■■■	中文：3. Prepare the ethanol series, 配好乙醇梯度
EN: Prepare 30%, 50%, 70%, and 100% EtOH in clearly labelled bottles. You will need these during dehydration, staining, and rehydration steps.	中文：先配好 30%、50%、70% 和 100% EtOH，并清楚标注。后面脱水、染色和复水步骤都会用到。
EN: Suggested small pilot volume: about 20 to 30 ml of each concentration for a small test batch, or more if your containers are large.	中文：建议的小规模预实验体积：如果只是少量样品，每个浓度先配约 20 到 30 ml 即可；如果容器较大，则需要适当增加。
EN: 4. Prepare storage solution if needed ■■■■■■■■■■	中文：4. Prepare storage solution if needed, 如需保存，再配保存液
EN: If your leaves cannot be processed immediately, prepare the formalin acetic acid storage solution before sampling.	中文：如果叶片采下后不能立即处理，请在取样前先配好福尔马林-醋酸保存液。
EN: 5. Prepare NaOH clearing solution ■■■■■■	中文：5. Prepare NaOH clearing solution, 配透明化溶液
EN: Prepare 2.5% NaOH for a gentler first trial, and 5% NaOH if needed for stronger clearing. Start with a small amount.	中文：建议先配 2.5% NaOH 作为较温和的预实验条件，如透明化不够，再准备 5% NaOH。第一次可以先少量配制。
EN: Suggested small pilot volume: about 20 to 30 ml per concentration.	中文：建议的小规模预实验体积：每个浓度先配约 20 到 30 ml。
EN: 6. Prepare bleach close to use time ■■■■■■■■	中文：6. Prepare bleach close to use time, 临用前准备漂白液
EN: Prepare the 50% bleach solution close to the time of use, rather than too early, so it stays effective.	中文：50% 漂白液建议在接近使用时再配，不要过早准备，这样更稳妥。
EN: Suggested small pilot volume: about 10 to 20 ml is often enough for a few leaves.	中文：建议的小规模预实验体积：如果只是几片叶子，通常先配 10 到 20 ml 就够。
EN: 7. Prepare staining solutions ■■■■■■	中文：7. Prepare staining solutions, 配染色液

EN: Prepare 1% safranin in 100% EtOH and 1% fast green in 100% EtOH. Because staining times may need optimisation, it is wise to start with small volumes.	中文：配好 1% 番红的 100% EtOH 溶液，以及 1% 固绿的 100% EtOH 溶液。由于染色时间通常需要优化，建议先小体积配制。
EN: Suggested small pilot volume: about 10 to 20 ml of each stain.	中文：建议的小规模预实验体积：每种染色液先配约 10 到 20 ml。
EN: 8. Set up reverse series and rinsing containers ■■■■■■■■■■	中文：8. Set up reverse series and rinsing containers, 准备复水和冲洗容器
EN: Before you start staining, make sure you already have the reverse alcohol series and water rinse containers ready. This avoids delays when the leaf is already stained.	中文：在开始染色前，最好先把反向乙醇梯度和清水冲洗容器也准备好。这样叶片染好后可以马上进入下一步，不会手忙脚乱。
EN: 9. Set up mounting and imaging materials ■■■■■■■■■■	中文：9. Set up mounting and imaging materials, 提前准备制片和成像材料
EN: Have CG5000 film, microscope setup, scanner, camera, and ImageJ ready before the leaf reaches the final stage.	中文：在叶片进入最后制片步骤前，就把 CG5000 胶片、显微镜、扫描仪、相机和 ImageJ 都准备好。
EN: 10. Prepare waste containers and safety items ■■■■■■■■■■	中文：10. Prepare waste containers and safety items, 准备废液容器和安全用品
EN: Set up separate waste containers according to your lab rules, and make sure gloves, lab coat, and eye protection are ready before handling formaldehyde, NaOH, and bleach.	中文：按照实验室规定准备好分类废液容器，并在接触甲醛、NaOH 和漂白剂之前确认手套、实验服和护目镜都已到位。
EN: Practical summary ■■■■	中文：Practical summary, 实操总结
EN: A good working order is: water, ethanol series, storage solution if needed, NaOH, bleach, stains, reverse series, then mounting and imaging setup. For a first barley pilot, small working volumes are usually the safest choice.	中文：一个比较顺手的准备顺序通常是：纯水，乙醇梯度，如有需要再配保存液，然后是 NaOH、漂液、染色液、反向梯度，最后准备制片和成像系统。对于第一次做大麦预实验，先用小体积工作液通常最稳妥。

Appendix 6. One page bench side workflow 一页式实验当天操作单

EN: This page is a simplified bench side guide for the actual experiment day. It is intended as a quick operational reminder, not as a replacement for the full protocol and laboratory safety rules.	中文：这一页是给实验当天使用的简化操作单，适合放在实验台旁边快速查看。它是提醒版，不替代完整 protocol 和实验室安全规定。
EN: Step 1. Sample collection ■■	中文：Step 1. Sample collection, 取样
EN: Collect labelled barley leaves. Record genotype, treatment, replicate, date, and leaf position or age.	中文：采集并标记好大麦叶片，记录基因型、处理、重复、日期，以及叶位或叶龄。
EN: Step 2. Storage if needed ■■■■	中文：Step 2. Storage if needed, 如需保存
EN: If leaves cannot be processed immediately, place them in the formalin acetic acid storage solution.	中文：如果不能立即处理叶片，将其放入福尔马林-醋酸保存液中。
EN: Step 3. Clearing ■■■	中文：Step 3. Clearing, 透明化
EN: Place leaves in 2.5% or 5% NaOH and monitor tissue softness regularly. Start with the gentler option when optimising barley leaves.	中文：将叶片放入 2.5% 或 5% NaOH 中，并定期观察组织是否变软。第一次优化大麦叶片时，建议先用较温和条件。
EN: Step 4. Rinse and bleach ■■■■	中文：Step 4. Rinse and bleach, 冲洗与漂白
EN: Rinse the cleared leaf, then if needed use 50% bleach briefly, usually only 10 to 20 seconds.	中文：透明化后先冲洗叶片，如有需要，再短时间使用 50% 漂液，通常仅 10 到 20 秒。
EN: Step 5. Ethanol series ■■■■	中文：Step 5. Ethanol series, 乙醇梯度
EN: Move the leaf through 30%, 50%, 70%, and 100% EtOH, up to about 5 minutes each step.	中文：将叶片依次转入 30%、50%、70% 和 100% EtOH 中，每步约不超过 5 分钟。
EN: Step 6. Safranin staining ■■■■	中文：Step 6. Safranin staining, 番红染色
EN: Stain in 1% safranin in 100% EtOH. Adjust the time according to leaf response and image quality.	中文：使用 1% 番红的 100% EtOH 溶液染色。根据叶片反应和图像质量调整时间。
EN: Step 7. Fast green staining ■■■■	中文：Step 7. Fast green staining, 固绿染色
EN: Stain in 1% fast green in 100% EtOH for only a few seconds.	中文：使用 1% 固绿的 100% EtOH 溶液染色，仅需数秒。
EN: Step 8. Reverse alcohol series ■■■■	中文：Step 8. Reverse alcohol series, 反向乙醇梯度
EN: Return the leaf through the reverse EtOH series back to water.	中文：将叶片通过反向乙醇梯度逐步回到水相。
EN: Step 9. Mounting ■■	中文：Step 9. Mounting, 制片
EN: Mount the leaf carefully on CG5000 transparency film or according to your local mounting method.	中文：将叶片小心制片到 CG5000 透明胶片上，或按你们实验室本地方法制片。
EN: Step 10. Imaging ■■	中文：Step 10. Imaging, 成像

EN: Scan the whole leaf using the original protocol setting of about 1,200 pixels per inch, or use a higher resolution if required by your local imaging system. Image vein details under the microscope, typically at 5x or 10x.	中文：整叶扫描可参考原 protocol 的约 1,200 pixels per inch，或根据本地成像系统需要使用更高分辨率。叶脉细节部分可在显微镜下拍摄，通常使用 5 倍或 10 倍物镜。
EN: Step 11. ImageJ analysis ImageJ ■■	中文：Step 11. ImageJ analysis, ImageJ 分析
EN: Measure leaf area, midrib length, second order vein length, higher order vein length, and free ending vein metrics as required.	中文：根据需要测量叶面积、中脉长度、二级叶脉长度、高级叶脉长度，以及自由末端叶脉相关指标。
EN: Step 12. Record and clean up ■■■■■	中文：Step 12. Record and clean up, 记录与收尾
EN: Write down actual treatment times, any problems, and image quality. Dispose of chemical waste according to lab rules.	中文：记录每一步的实际处理时间、出现的问题和图像质量，并按实验室规定处理化学废液。
EN: Quick reminders ■■■■	中文：Quick reminders, 快速提醒
EN: Do not over clear in NaOH. Do not over bleach. Do not over stain in safranin. Handle softened barley leaves gently during transfers. Keep your timing records carefully for later optimisation.	中文：不要在 NaOH 中处理过度，不要漂白过度，也不要让番红染色过深。大麦叶片在软化后转移时要特别轻柔。请认真记录时间，便于后续优化。